Mobin Azimipanah

EU work authorization · no visa sponsorship required

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Education

Polytechnic University of Milan, Milan, Italy — Sep 2022 – Apr 2025
M.Sc. in Electrical Engineering - Computer Science (Thesis:Multi-RAG Framework with Prof.Carman, Mark James)

University Grenoble Alpes, Grenoble, France — Sep 2023 – Sep 2024
M.Sc. in Electrical Engineering - Computer Science - Machine Learning and Robotics

Urmia University, Urmia, Iran — Sep 2017 – Sep 2021
B.Sc. in Electrical Engineering - Power Electronics, GPA: 3.8/4

Achievements

- Ranked top 1% (2,000 of 300,000) in Iran's National University Entrance Exam (Math/Physics).
- Ranked top 3 in B.Sc. Electrical Engineering cohort; full scholarship for academic excellence.

Work Experience

Hermes Corporate, Florence, Italy — AI Engineer (Full-time) — June 2025 – Present
- Consulting Baker Hughes on AI infrastructure: OpenAPI and MCP services for surrogate model training, DOE generation, preprocessing, and reporting. Stack: FastAPI, PydanticAI, PostgreSQL, MLflow, Docker, NGINX; deployed on AWS.

Canon Research Center France, Rennes, France – Intern — Apr 2024 – Oct 2024
- Led the development of an automated RAG system to identify relevant matches between IP documents, achieving over 90% coverage of the IP content and its references in under 3 minutes—took researchers days to complete.
- Designed hierarchical ensemble retrievers and utilized AutoGen with GraphRAG for multi-agent workflows.
- Fine-tuned and optimized local LLMs using LoRA, MoRA, VeRA, and quantization methods.

Freelancer at Outlier, Remote – AI Expert & Prompt Engineering — Jan 2025 – Present
- Delivered +50 innovative prompt engineering strategies for diverse client projects to help train frontier models.

Selected Projects

Zirak.dev[Link] — January 2025 – Present
- Engineered a full-stack AI assistant platform (Next.js, React) featuring a custom Python-based agentic workflow that autonomously decomposes complex multi-step tasks and manages completion loops. (Coming Soon)

2024 University of California, Berkeley Large Language Model Agent Hackathon[Link] — Nov 2024 – December 2024
- Led technical architecture and development of regulAItor, a multi-agent LLM application leveraging AutoGen to analyze 1,000+ FDA warning letters, streamlining pharmaceutical regulatory compliance through automated analysis.

Woging.com – Multi-LLM Chatbot for Pet Care (Team Member) — Oct 2024 – January 2025
- Built a Multi-LLM chatbot using AutoGen, RAG, and MongoDB for pet owner and veterinarian support.

Cryptocurrency Trading and Prediction System — *Jun 2024 – Sep 2024*
- Developed a Dockerized system for real-time crypto trading and Bitcoin prediction using Kafka, InfluxDB, and xLSTM models (1M+ data points) with 72.87% test accuracy, plus TensorBoard and visualizations.

LLM, Vision-Text & Computer Vision Coursework — Jan 2023 – Jan 2024
- Built a vision-text transformer for image captioning; applied X-CLIP for video-text probability ranking; designed regularized GRU/Conv/LSTM models for sentiment analysis on Sentiment140.
- Implemented CNN classifiers (TensorFlow) and Faster R-CNN object detection on Cats vs Dogs, Sign Language MNIST, and COCO datasets.

Skills

ML/AI: PyTorch, TensorFlow, Transformers, LangChain, Llama-index, AutoGen, GraphRAG, MCP, PydanticAI, MLflow, LoRA/quant
Backend & Data: Python, FastAPI, PostgreSQL, MongoDB, Kafka, InfluxDB, Databricks
Frontend & Infra: Next.js, React, Docker, AWS (ECS, S3), NGINX, Git

Certifications

Large Language Models: Foundation Models from the Ground Up (edX) · Sequences, Time Series and Prediction (DeepLearning.AI) · TensorFlow on Google Cloud (Google Cloud) · Natural Language Processing in TensorFlow (DeepLearning.AI) · Computer Vision and Image Processing Essentials (IBM) · Convolutional Neural Networks in TensorFlow (DeepLearning.AI) · Machine Learning Specialization (Stanford University)

Research & Internships & Activities

Publication: Open Circuit Fault Detection for Seven-Level Hybrid Inverters. Presented at the *2021 29th Iranian Conference on Electrical Engineering (ICEE)*, IEEE, Tehran, Iran. [Link]

Research Assistant — Politecnico di Milano
Sparse Techniques in Neural Network Backpropagation — Feb 2023 – May 2025
Under Prof. Agosta, focused on improving the efficiency of sparse techniques in neural network training.

Activities *Hackathon Politecnico di Milano (November 2024), Industry 4.0 Lab (Politecnico di Milano, November 2022)*